Exponentials & Logarithms

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

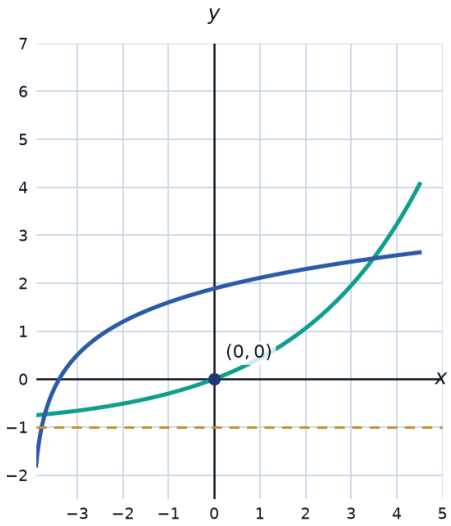
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $\ln y = 2\ln x + \ln 3$, express y in terms of x. Hence sketch the graph.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

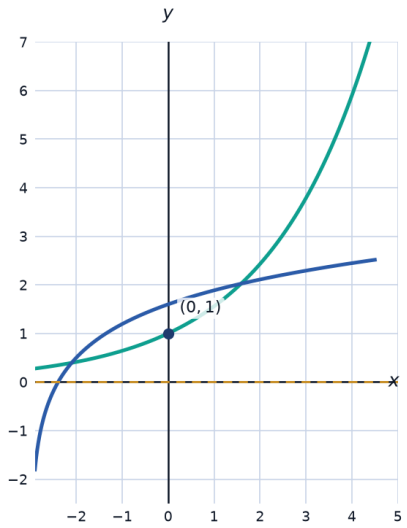
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

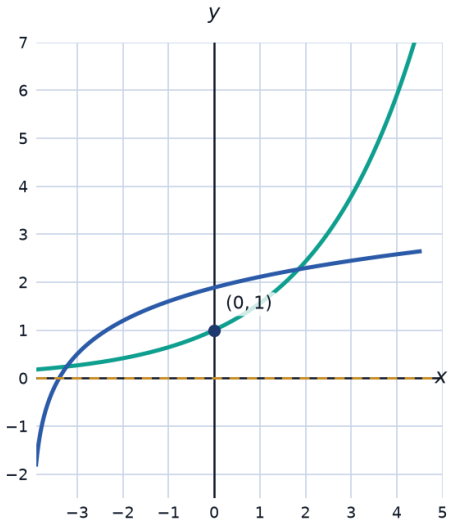
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

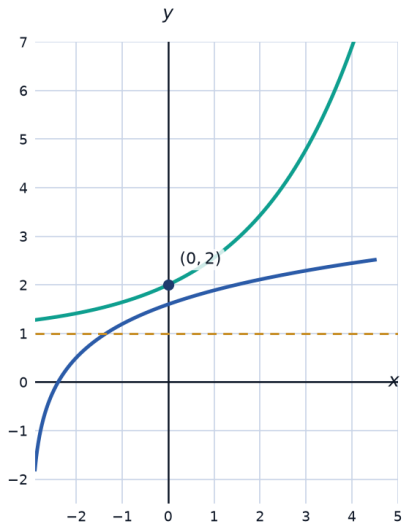
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Advanced Trigonometry

Solve the separable DE $\frac{dy}{dx} = \frac{x+1}{y}$ given $y = 2$ when $x = 0$.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Differentiation

Given that $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$, find $\frac{z_1}{z_2}$ in the form $a + bi$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Integration

A curve is defined parametrically by $x = t^2 + 1$, $y = 2t^3 - 3t$. Find $\frac{dy}{dx}$ in terms of t and the value at $t = 1$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Sequences & Series

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Vectors

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$ where $R > 0$ and $0 < \alpha < 90°$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the equation $e^{2x-1} = 5$, giving your answer in exact form.

a. $x = \frac{1 + \ln 5}{2}$

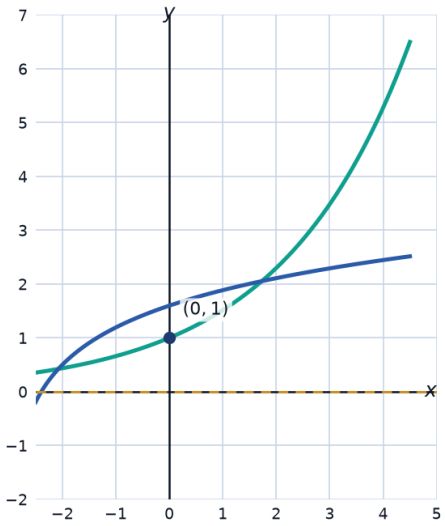
b. $x = \frac{\ln 5 - 1}{2}$

c. $x = \ln\left(\frac{5}{2}\right) + 1$

d. $x = \frac{\ln 6}{2}$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the expression $e^{3\ln x} + \ln(e^{2x})$ for $x > 0$.



a. $x^3 + 2x$

b. $3x + 2x = 5x$

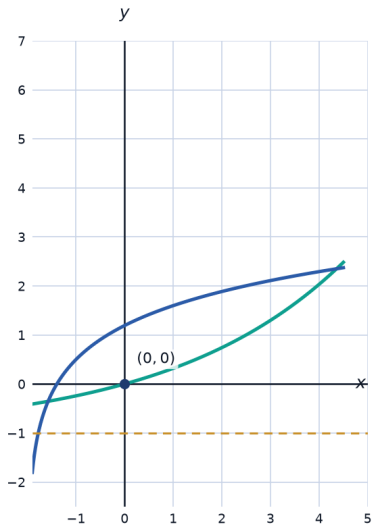
c. $x^3 + e^{2x}$

d. $3x^2 + 2$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the equation $2^{x+1} = 3^x$, giving x in terms of natural logarithms.



a. $x = \frac{\ln 2}{\ln 3 - \ln 2}$

b. $x = \frac{\ln 3}{\ln 3 - \ln 2}$

c. $x = \frac{\ln 2}{\ln 3 + \ln 2}$

d. $x = \frac{1}{\ln(3/2)}$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the equation $e^x - 6e^{-x} = 1$ for real x.

a. $x = \ln 3$

b. $x = \ln 3$ or $x = \ln(-2)$

c. $x = \ln 2$

d. $x = 3$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $5^{2x-1} = 7$, giving your answer correct to 3 significant figures.

a. 1.10

b. 0.852

c. 1.21

d. 0.945

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The variables x and y satisfy $y = ab^x$. When $\ln y$ is plotted against x, a straight line is obtained. What are the gradient and vertical intercept?

a. Gradient $= \ln b$, Intercept $= \ln a$

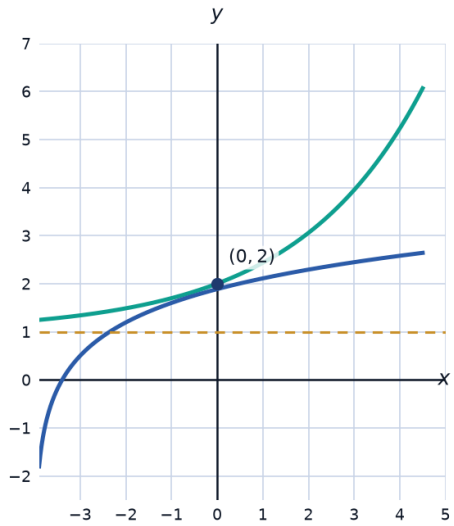
b. Gradient $= b$, Intercept $= a$

c. Gradient $= \ln a$, Intercept $= \ln b$

d. Gradient $= a$, Intercept $= \ln b$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The variables x and y satisfy $y = kx^n$. The line of $\ln y$ against $\ln x$ has gradient 2.5 and intercept $\ln y = 1.386$. Find k and n.



a. $k = 4, n = 2.5$

b. $k = 2.5, n = 4$

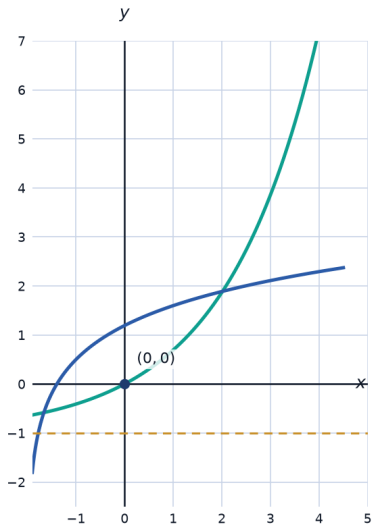
c. $k = 1.386, n = 2.5$

d. $k = e^{2.5}, n = 1.386$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Points $(2, 3.2)$ and $(5, 5.6)$ lie on the straight line of $\ln y$ against x. Find the value of b if $y = ab^x$.



a. $b = e^{0.8} \approx 2.226$

b. $b = 0.8$

c. $b = 2.4$

d. $b = e^{1.6} \approx 4.95$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The mass of a radioactive substance decays according to $M(t) = 80e^{-0.05t}$, where t is in years. Find the half-life of the substance in years.

a. ≈ 13.9 years

b. 20.0 years

c. 10.0 years

d. 15.5 years

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bacterial culture grows according to $N(t) = 500e^{kt}$. If $N(3) = 2000$, find the exact value of k.

a. $k = \frac{\ln 4}{3}$

b. $k = \frac{\ln 2}{3}$

c. $k = \frac{4}{3}$

d. $k = \ln\left(\frac{4}{3}\right)$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A hot beverage cools according to Newton's law of cooling: $T(t) = 20 + 70e^{-0.08t}$, where T is temperature in $^\circ$C and t is time in minutes. How long does it take to cool to $45\,^\circ$C?

a. ≈ 12.9 min

b. 15.2 min

c. 10.5 min

d. 18.0 min

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Given $\log_b 3 = A$, $\log_b 11 = B$ and $\log_b 5 = C$, express $\log_b 25$ in terms of A, B and C.

a. A+2C

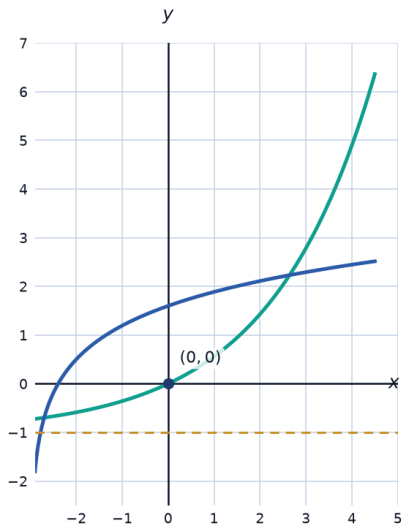
b. 2A

c. 3C

d. 2C

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $\log_b 3 = A$ and $\log_b 2 = B$, express $\log_b 3$ in terms of A and B.



a. A+B

b. A

c. 2A

d. B

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $\tan\theta = -\frac{6}{5}$ with θ in Quadrant II, state the sign of $\sec\theta$.

a. positive

b. negative

c. undefined

d. zero

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An arithmetic sequence has first term $a = 1$ and common difference $d = -4$. Find the sum of the first 3 terms.

a. -10

b. -7

c. -13

d. -9

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\tan\theta = 3$ with θ in Quadrant I, state the sign of $\sec\theta$.

a. undefined

b. zero

c. positive

d. negative

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric series has first term $a = 2$ and common ratio $r = -\frac{1}{4}$. Find the sum to infinity.

a. $\frac{8}{5}$

b. $\frac{18}{5}$

c. $\frac{5}{2}$

d. $-\frac{8}{5}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Sequences & Series

Find the remainder when $x^2 - 5x - 4$ is divided by $(x - 3)$.

a. -2

b. 10

c. -10

d. -9

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Vectors

The locus $|z - (3 - 2i)| = 4$ is a circle. What is its center?

a. (3, -1)

b. (3, -2)

c. (4, -2)

d. (-3, 2)

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Given $y = 5e^{2x}$, find the coefficient of e^{2x} in $\frac{dy}{dx}$.

a. 11

b. 10

c. 2

d. 7

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Complex Numbers

Solve $\cos(\theta) = -\frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the smallest solution (in degrees).

a. 510

b. 210

c. 151

d. 150

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the point of intersection (with positive imaginary part) of $|z| = 5$ and $\mathrm{Re}(z) = 3$.

a. (3, -4)

b. (4, 3)

c. (3, 4)

d. (5, 3)

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Numerical Methods

Solve $\sin(\theta) = \frac{\sqrt{3}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. Find the largest solution (in degrees).

a. 119

b. 0

c. 60

d. 120

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Differential Equations

A geometric series with common ratio $r = \frac{1}{3}$ has sum to infinity $S_\infty = 12$. Find the first term a.

a. 11

b. 7

c. 8

d. -8

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 12.01 Poisson Distribution

Find the magnitude of the vector $\mathbf{v} = (4, -4, 7)$.

a. 15

b. 9

c. 8

d. 10

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